

Diego Pestana

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Education

University of Kansas – M.S. in Computer Science

Aug 2025 - May 2027

- Coursework: Machine Learning, Reinforcement Learning, Information Retrieval

Universidad Metropolitana – B.S. in Systems Engineering

Sept 2019 - Feb 2024

- Honors: summa cum laude, **GPA: 18.52/20.00**
- Coursework: Computer Architecture, Data Structures, Databases, Operating Systems

Experience

Software Engineer II, Venezolano de Crédito (Venezuela's Largest Corporate Bank)

Aug 2023 – July 2025

- Built **tailored banking features** for large clients, supporting client acquisition, retention, and vertical growth
- Designed a scalable RESTful API using **Java**, **Spring Boot** and **SQL** that securely allows outside parties to access bank's financial services, facilitating **10,000+** users and supporting **50,000+** daily operations
- Developed and implemented a supervised financial transaction classification model using a bag of words and random forest, enabling personalized financial recommendations and targeted marketing campaigns
- Led a 5-person technical team in delivering continuous features and updates to a **Flutter** mobile app, meeting user needs and driving a **30%** increase in store reviews that boosted the app's rating
- Integrated with a third-party tax administration service to provide online tax payments through a full-stack web app made using **Java** and **ReactJS**, resulting in **\$500,000+** collected thus far
- Built a **JavaScript** library that generates and scans QRs with users' public banking information, easing the exchange of data and increasing by **10%** the amount of daily P2P operations

Software Engineer I, Venezolano de Crédito (Venezuela's Largest Corporate Bank)

Jan 2023 – July 2023

- Optimized query execution times of an **Oracle** database supporting multiple internal systems by partitioning large tables and creating materialized views, improving the systems' responsiveness and eliminating timeouts
- Refactored a **Java** legacy codebase using a **TDD** approach, reducing tight coupling and increasing adaptability
- Migrated **5,000+** lines of **JQuery** to **JavaScript** in order to improve compatibility with modern frameworks
- Wrote **Python** scripts to generate daily currency exchange reports, minimizing manual intervention

Software Engineer Intern, Venezolano de Crédito (Venezuela's Largest Corporate Bank)

Sept 2022 – Dec 2022

- Redesigned **62** non-responsive pages with **CSS**, making them fit correctly on different screen sizes, e.g. mobile
- Solved **30+** customer issues reported by support tickets, boosting web app user satisfaction scores by **15%**

Research & Projects

Automating Credit Card Limit Adjustments Using Machine Learning

arxiv.org/abs/2501.10451

- Collected a dataset of **10,000+** historical credit card limit adjustment decisions using **SQL** for data extraction and **PyTorch** for preprocessing, serving as the training data for the models
- Trained and compared multiple **ML** models (**MLP** and **XGBoost**) using a **cost-sensitive** learning approach, considering the trade-off between accuracy and interpretability to satisfy regulatory needs
- Optimized model hyperparameters using grid search with **10-fold cross-validation**, resulting in a final model that achieved almost perfect agreement with human expert decisions, as measured by Cohen's kappa coefficient
- Presented research findings and results to business areas, driving the adoption and implementation of the model

Technical Skills

Languages: Java, Python, JavaScript, SQL, Dart, C++, Linux Shell Script

Technologies: Spring Boot, PyTorch, ReactJS, Oracle, PostgreSQL, Flutter, Git

AI & Agents: Claude Code, MCP, Agent Orchestration, Prompt Engineering

Achievements & Awards

- Secured a **Full Academic Scholarship** totaling **\$70,000** at the University of Kansas
- Awarded **Dean's List** Recognition in 2022 and 2023 at Universidad Metropolitana